

MINJIE MAO

Software Development Engineer

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EDUCATION

University of Southern California	Los Angeles, USA
Master of Science in Computer Science GPA - 3.81/4.00	Sep 2023 – May 2025
University of Michigan - Ann Arbor	Ann Arbor, USA
Bachelor of Science in Computer Science (Dual Degree) GPA - 3.78/4.00	Sep 2021 – Apr 2023
Shanghai Jiao Tong University	Shanghai, China
Bachelor of Science in Electrical and Computer Engineering (Dual Degree) GPA - 3.62/4.00	Sep 2019 – Aug 2023

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, SystemVerilog, JavaScript, TypeScript, Java, Kotlin, Bash Scripting
Frameworks & Development Tools: Pytorch, Numpy, DSPy, AWS CDK, MLflow, NodeJS, React, Angular
Infra & Service: Elastic Kubernetes Cluster, Elastic Container Service, Bedrock, Cloudwatch, FSx

WORK EXPERIENCE

Amazon Web Services	Bellevue, USA
Software Dev Engineer - GenAI for Kernels Bedrock, Kubernetes, DSPy, MLflow	June 2025 – Present
- Configured an Elastic K8s cluster on Trainium nodes to benchmark LLM-generated kernel performance. - Architected and benchmarked two multi-agent systems, including instruction-guided RAG agents and self-evolving agents, resulting in a 1.23x average kernel speedup for the superior agentic workflow. - Deployed an MLflow server on ECS with RDS & S3 backends to analyze optimization trajectories.	
Amazon Web Services	San Jose, USA
Software Dev Engineer Intern - Machine Learning Chip Accelerator LLM, torchXLA	May 2024 – Aug 2024
- Implemented multi-modal model inference on AWS ML chips (Trn1/Inf2) under NeuronxDistributed framework. - Enabled input and output channel level tensor parallel convolutional layer on neuron device. - Designed preprocessing of the embedding mask on CPU to avoid tensor materialization and computation graph truncation.	

PROJECTS

Reliable Transport Protocol & BBR Congestion Control C/C++, Socket Programming, Quagga, BBR	Sep 2023 – Dec 2023
- Built a reliable transport protocol on top of UDP with cumulative acknowledgment and sliding window to resolve issues of packet loss, delay, corruption, duplication, and reordering. - Configured OSPF, iBGP and eBGP within an Autonomous System using Quagga routing suite. - Added Bottleneck Bandwidth and Round-trip propagation time(BBR) congestion control protocol to simplified TCP.	
R10K-style Out-of-Order Superscalar Processor SystemVerilog	Jan 2023 - Apr 2023
- Oversaw the intricate design of critical components of an R10k-style out-of-order superscalar processor, including the Reorder Buffer (ROB), Reservation Stations (RS), and Physical Register File (PRF) to increase instruction level parallelism. - Implemented a multiport non-blocking instruction cache and an instruction fetcher coordinating with branch predictor. - Engineered a 2-way set-associative data cache featuring a victim cache under LRU policy, to minimize cache misses.	
Optimization of the Neural Network Convolutional Forward Pass CUDA C/C++	Sep 2022 – Dec 2022
- Adopted shared memory tiling technique to reduce global memory traffic of the forward path of convolutional layers. - Reduced convolution kernel to a highly efficient matrix multiplication kernel with input unfolding and replicating. - Exploited multiple fine-tuned kernels tailored for various convolutional layers within deep learning models.	
Operating System Emulation C/C++, Multithreading, Network File System	Sep 2022 – Dec 2022
- Customized a thread library with mutex support to manage uniprocessor scheduling. - Implemented virtual memory management pager, optimizing memory allocation and resource utilization. - Orchestrated emulation of a multi-threaded, multi-client, and secure network file system server, minimizing disk I/Os.	
Responsive Web & Android App Dev Typescript, NodeJS, Angular, MongoDB, Kotlin, REST API	Jan 2024 – May 2024
- Developed a real-time stock search website with responsive design using Bootstrap and Angular framework. - Deployed NodeJS backend server and MongoDB Atlas to persistently store user information. - Enabled gesture control features on migrated android stock search application.	